Drugs

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Drugs [1]

Which statement describes a drug?

A any substance that kills microorganisms inside the body

B any substance made by the body that controls chemical reactions in cells

C any substance taken into the body that modifies or affects chemical reactions in the body

D any substance taken into the body that is used to provide energy or materials for growth

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2 · Part 1 Core Drugs [1]

Four statements about drugs are listed.

1 Every drug is a medicine prescribed by a doctor.

2 Antibiotics are drugs.

3 To act as a drug, a substance must be taken into the body.

4 Every drug kills cells in the body.

Which statements are correct?

A 2 and 3

B 1 and 2

C 1 and 4

D 3 and 4

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3 · Part 1 Core Drugs [1]

Which row shows the effect of antibiotics on bacteria and on viruses?

	effect on bacteria	effect on viruses
A	killed	killed
B	not affected	killed
C	not affected	not affected
D	killed	not affected

4 · Part 1 Core Drugs [1]

Common colds are caused by viruses. Ravi has a cold and asks his doctor for an antibiotic. The doctor says that an antibiotic will not cure his cold.

Why is the doctor correct?

A The cold virus would quickly become resistant to the antibiotic.

B Antibiotics do not affect viruses.

C The antibiotic would kill the white blood cells that are fighting the virus.

D Colds are not transmissible diseases.

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5 · Part 1 Core Drugs [1]

Penicillin is an antibiotic. Four patients, P, Q, R and S, each have an infection. The table gives information about the pathogen causing each infection.

patient	type of pathogen	pathogen resistant to penicillin
P	bacterium	no
Q	bacterium	yes
R	virus	not tested
S	virus	not tested

For which patient is penicillin most likely to cure the infection?

A P

B Q

C R

D S

6 · Part 1 Core Drugs [1]

Some bacteria are resistant to an antibiotic.

What is the result of this resistance?

A The antibiotic becomes effective against viruses as well as bacteria.

B The antibiotic kills the resistant bacteria more quickly.

C The antibiotic becomes less effective at treating infections caused by these bacteria.

D The resistant bacteria are no longer able to cause disease.

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7 · Part 1 Core Drugs [1]

The surface of a dish of agar jelly was covered evenly with one type of bacterium. Four small paper discs were placed on the agar. Each disc had been soaked in a different antibiotic, W, X, Y or Z. The dish was kept warm for two days.

The bacteria grew over the agar, except in a circular clear area around some of the discs. The table shows the diameter of the clear area around each disc.

antibiotic	diameter of clear area / mm
W	0
X	24
Y	11
Z	6

Which antibiotic was most effective at killing this type of bacterium?

A W

B X

C Y

D Z

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8 · Part 1 Extended Drugs [1]

Doctors are advised to prescribe antibiotics only when they are essential.

Which statement explains how this limits the development of resistant bacteria such as MRSA?

A Antibiotics cause the mutations that make bacteria resistant, so using fewer antibiotics causes fewer mutations.

B The human body becomes resistant to antibiotics that are used too often.

C Resistant bacteria are killed when antibiotics are not being used.

D Bacteria are exposed to antibiotics less often, so resistant bacteria are selected less often.

9 · Part 1 Extended Drugs [1]

MRSA is a strain of bacteria that is resistant to several antibiotics. Four events in the development of a population of resistant bacteria are listed.

1 Bacteria with the gene for resistance survive and reproduce.

2 A random mutation gives one bacterium a gene for resistance.

3 An antibiotic is used and kills the bacteria that are not resistant.

4 The gene for resistance is passed on until most of the population is resistant.

What is the correct order of these events?

A 3 → 2 → 1 → 4

B 2 → 1 → 3 → 4

C 2 → 3 → 1 → 4

D 3 → 1 → 2 → 4

10 · Part 1 Core Drugs [1]

Scientists tested samples of one species of bacterium from hospital patients. They recorded the percentage of samples that were resistant to antibiotic T.

year	samples resistant to antibiotic T / %
2000	4
2005	9
2010	17
2015	28
2020	41

Which conclusion is supported by the data?

A Antibiotic T was likely to be less effective at treating infections caused by this bacterium in 2020 than in 2000.

B All of the samples of this bacterium were resistant to antibiotic T by 2020.

C The percentage of resistant samples doubled every five years.

D The number of patients infected with this bacterium increased between 2000 and 2020.

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11 · Part 1 Core Drugs [4]

Antibiotics are one type of drug.

(a) Describe what is meant by a drug. [2]

(b) State the type of pathogen that antibiotics are used to treat. [1]

(c) Influenza is caused by a virus. State why antibiotics are not used to cure influenza. [1]

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12 · Part 1 Core Drugs [5]

Two flasks of nutrient broth were each given the same number of one type of bacterium. An antibiotic was added to flask 1. Nothing was added to flask 2. Both flasks were kept at 30 °C.

Every 2 hours, a sample was taken from each flask and the number of live bacteria in 1 cm^3 of broth was found. The results are shown in the table.

time / h	**number of live bacteria in 1 cm^3 of flask 1 (antibiotic)**	**number of live bacteria in 1 cm^3 of flask 2 (no antibiotic)**
0	2000	2000
2	1200	4000
4	600	8000
6	300	16 000
8	100	32 000

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(a) Describe the effect of the antibiotic on the bacteria. Use data from the table in your answer. [2]

(b) Calculate the percentage decrease in the number of live bacteria in flask 1 between 0 hours and 8 hours. [2]

(c) Explain why flask 2 was included in the investigation. [1]

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13 · Part 1 Core Drugs [4]

Three patients visit a clinic.

- Patient 1 has pneumonia caused by a bacterium.
- Patient 2 has influenza caused by a virus.
- Patient 3 has HIV, which is a virus.

(a) State which patient would be helped by being given an antibiotic. [1]

(b) Patient 2 asks for an antibiotic. Explain why the doctor refuses. [2]

(c) Some months later, patient 3 develops a lung infection caused by bacteria. Suggest why the doctor now prescribes an antibiotic for patient 3. [1]

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14 · Part 1 Core Drugs [5]

Samples of one type of bacterium were collected from 200 patients with urinary infections. Each sample was tested to find out whether the bacteria were resistant to three antibiotics, E, F and G.

antibiotic	number of samples tested	number of samples resistant
E	200	18
F	200	126
G	200	2

(a) Calculate the percentage of samples that were resistant to antibiotic F. [2]

(b) State which antibiotic would be least effective for treating these infections. Explain your answer using the data. [2]

(c) State which antibiotic a doctor should choose to treat a new patient with this infection. [1]

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15 · Part 1 Core Drugs [6]

Doctors studied patients with sore throats. Some sore throats were caused by bacteria and some were caused by viruses. Half of each group was given an antibiotic and half was given no antibiotic. The doctors recorded how many patients recovered within 7 days.

cause of sore throat	treatment	number of patients	number recovered within 7 days
bacteria	antibiotic	80	72
bacteria	no antibiotic	80	36
viruses	antibiotic	60	33
viruses	no antibiotic	60	33

(a) Calculate the percentage of patients with a bacterial sore throat given an antibiotic who recovered within 7 days. [2]

(b) Compare the effect of the antibiotic on patients with bacterial sore throats and on patients with viral sore throats. [2]

(c) Explain the results for the patients with viral sore throats. [2]

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16 · Part 1 Core Drugs [5]

Maria had a wound on her leg that became infected with bacteria. She took an antibiotic every day. Each time the wound was checked, a swab was taken and the bacteria on it were counted.

day	number of bacteria on swab
0	5000
2	1500
4	400
6	900
8	2400

(a) Describe the changes in the number of bacteria between day 0 and day 8. [2]

(b) Maria took the antibiotic every day throughout the 8 days. Suggest an explanation for the change in the number of bacteria between day 4 and day 8. [2]

(c) On day 8 the doctor changed the antibiotic to a different one. State why. [1]

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17 · Part 1 Core Drugs [4]

State whether each statement is true or false.

(a) Antibiotics are used to treat infections caused by bacteria. [1]

(b) Antibiotics can be used to cure a cold caused by a virus. [1]

(c) Some bacteria are resistant to antibiotics. [1]

(d) Every drug is a substance made inside the body. [1]

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18 · Part 1 Core Drugs [4]

People in a town were asked whether they think antibiotics will cure a cold. The table shows the percentage in each age group who answered yes.

age group / years	people who think antibiotics will cure a cold / %
16–30	38
31–50	30
51–70	22
over 70	26

(a) State the age group with the highest percentage of people who think antibiotics will cure a cold. [1]

(b) There were 400 people in the 16–30 age group. Calculate how many of them think antibiotics will cure a cold. [1]

(c) Colds are caused by viruses. Explain why the people who answered yes are wrong. [2]

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19 · Part 1 Extended Drugs [6]

MRSA is a strain of the bacterium *Staphylococcus aureus* that is resistant to several antibiotics.

(a) Most of the bacteria in a population of *S. aureus* are killed by an antibiotic, but a few have a gene that makes them resistant. Describe what happens to this population when the antibiotic is used many times. [3]

(b) Doctors in a hospital are told to prescribe antibiotics only when they are essential. For example, they do not give antibiotics for mild infections that the body can fight off without help. Explain how this limits the development of resistant bacteria such as MRSA. [2]

(c) State why infections caused by MRSA are difficult to treat. [1]

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20 · Part 1 Extended Drugs [5]

In one country, doctors were asked to reduce the number of antibiotic prescriptions they wrote. The table shows the number of antibiotic prescriptions and the percentage of *Staphylococcus aureus* infections that were caused by MRSA.

year	antibiotic prescriptions per 1000 people	*S. aureus* infections caused by MRSA / %
2010	980	42
2012	900	39
2014	760	33
2016	640	26
2018	600	22

(a) Calculate the percentage decrease in antibiotic prescriptions per 1000 people between 2010 and 2018. Give your answer to three significant figures. [2]

(b) Describe the relationship between the number of antibiotic prescriptions and the percentage of infections caused by MRSA. [1]

(c) Suggest one change in hospitals, other than fewer prescriptions, that could also have helped to reduce MRSA infections. [1]

(d) Doctors were not told to stop prescribing antibiotics completely. Suggest why. [1]

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